

Accepted Service Adaptation: Continuation Prices, Transfer Coordinates, and Certified Gains

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Abstract. A service provider gains from adaptation only if each promised continuation remains acceptable. We study the value of expanding an optimized restricted contract under continuation participation, payment commitments, and switching costs. Our principal result characterizes the entire value frontier as contractual information restrictions are progressively released. On a regular segment, restriction prices determine the initial gain, a reduced curvature matrix determines its decay, and explicit continuation-slack ratios identify the first contractual bottleneck. Reoptimizing both classes yields a second distinction: greater continuation capacity can reduce the adaptation premium, whereas greater switching friction can increase it. Exact examples and a rent decomposition identify both effects. A promised-payment recursion establishes when public-state contracts can be implemented without storing the full history tree and when this compression fails. We evaluate fixed, time-only, current-regime, limited-memory, and full-history restrictions on common instances. Reusable restricted-value envelopes support economic certification without solving a new comparator at every nearby reward query. Radius-refined diagnostics explain the local behavior of robust outer comparators. Historical learned implementations and their unfavorable certified timing comparisons remain available as implementation evidence, separate from the contractual value results.

Keywords: continuation participation; service adaptation; contractual flexibility.

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Area of review: Stochastic Models.

1 Introduction

A service provider can find a more profitable operating policy and still be unable to offer it. A customer with an exit right evaluates the promised service and transfers remaining at the current review, not only the surplus forecast when the agreement was signed. An attractive aggregate adjustment can therefore fail in a particular continuation. An operational improvement must respect these continuation commitments as well as physical capacity and the cost of changing service tiers.

The relevant comparison is with the best implementable restricted contract. A time-only amendment, for example, may be appropriate when a service organization can publish a calendar of adjustments but cannot maintain a separate rule for every observed regime. A current-regime amendment permits more information. A limited-memory amendment permits more still. Comparing an adaptive policy with an arbitrary fixed protocol would mix the value of additional flexibility with the cost of having selected the original protocol poorly. We optimize both sides of each comparison.

This distinction changes the comparative statics. Relaxing a continuation cap improves each optimized value, but it need not improve their difference. A restricted contract may benefit more from the relaxation than an already adaptive one. Conversely, a restriction can force costly changes that the full class can avoid; increasing switching friction can then increase the value of removing the restriction. We give exact examples, rather than infer the adaptation premium from a comparative static for only one of the two values.

Our main structural result describes a controlled release of contractual restrictions. Starting from an optimized restricted contract, it identifies the marginal value of permitting unequal decisions at previously pooled histories, the curvature of this value, and the first continuation constraint that exhausts its slack. The result gives an exact policy path until the first change in the active contractual constraints. Successive segments describe the complete release frontier. It also gives the smallest release needed to attain a specified gain on the initial segment. The underlying parametric optimization tools are established; the result identifies the operational objects that must be priced and the continuation bottlenecks that limit an accepted improvement.

A second contribution is the comparator-adjusted rent decomposition. It attributes changes in the adaptation premium to the difference between full-class and restricted-class continuation prices. It separates the value of adding information from the value of relaxing participation or capacity. A hierarchy of optimized contracts then assigns each increment of value to a specified expansion of the implementable menu. These increments are conditional on the ordering of the restrictions, not an invariant allocation of value among information features.

We make the exit-review institution primary. Public demand, fulfillment, service tiers, and the transfer schedule are contractible; there is no hidden effort or private type. A promised-payment state summarizes the remaining participation obligation when the primitives and outside opportunities have a finite Markov description. This is an augmented state: observing the current demand regime alone does not reveal the inherited tier or the remaining payment promise. The recursion is exact under its stated sufficiency conditions. It does not turn an exponentially growing history tree into a universally polynomial long-horizon problem.

Finally, a policy certificate and an economic certificate answer different questions. The first checks whether an implemented decision is accepted and sufficiently close to the full optimum. The second checks

whether its value exceeds an optimized restriction. We provide a reusable upper envelope for the latter task when only rewards vary. We also explain why the robust comparator used in the earlier computational record converges at small uncertainty radii despite a steep initial slack curve. These results support deployment accounting; they do not replace the contractual contribution with a claim about a particular prediction architecture.

The numerical study follows this hierarchy. On the same legacy instances we compare fixed, time-only, regime-dependent, two-period-memory, and full-history amendments. A separate public-Markov family compares actual service tiers rather than amendments to heterogeneous legacy tiers. All comparisons retain the same accepted set and use independently recomputed rational bounds. We retain the earlier finding that the tested learned routes do not amortize against lifted classical continuation under the matched certified protocol. No computational advantage for learning is needed for the structural value results.

2 Relation to Dynamic Contracting and Optimization

Continuation states are established in dynamic contracting. Spear and Srivastava (1987) and Thomas and Worrall (1988) use recursive contractual promises in environments with incentive or enforcement constraints. In Operations Research, Chen et al. (2020) study optimal monitoring when unobserved effort changes the arrival of adverse events. Their formulation uses the agent’s promised continuation utility as a state, with promise keeping, incentive compatibility, and continued participation determining its evolution. Liang et al. (2023) study resource-allocation contracts that induce effort to reduce adverse events and derive implementable allocation structures using dynamic incentives and self-generating sets.

Our public-state institution does not solve those information problems. Its controls are contingent service tiers under externally fixed prices, and its binding constraints are continuation participation and physical or payment commitments. The question is which accepted deviations improve upon an already optimized information restriction. A *continuation price* here is a multiplier on a continuation participation constraint. It is not the customer’s promised utility. The promised payment in Section 6 is instead a primal state variable. This distinction prevents a dual representation from being mistaken for a new promised-utility construction.

Constrained Markov decision processes supply resource-state and occupation-measure methods (Altman, 1999). Strong convexity, normal cones, and sensitivity of convex values are standard (Boyd and Vandenberghe, 2004). Absolute-value switching is an instance of a generalized-lasso penalty (Tibshirani and Taylor, 2011). Explicit parametric quadratic programming supplies the piecewise-affine policy and piecewise-quadratic value structure used below (Bemporad et al., 2002). We do not claim any of these general principles as new. Our use of them distinguishes restriction rents from participation rents, identifies the continuation-slack event that terminates a profitable release, and accounts for reoptimization of the comparator. The transfer identity by itself establishes none of those value comparisons.

Robust set containment is also an established device (Ben-Tal and Nemirovski, 2000; Bertsimas and Sim, 2004). We use it only for certification and quantify its local loss. Numerical optimizers, including OSQP (Stellato et al., 2020), produce candidate decisions and prices. The evidence below accepts a decision only after a separate rational calculation has checked feasibility and its primal–dual bracket. The historical

value-gradient and learned-price results remain in the accompanying research archive, with their assumptions and proofs intact; they are implementation results rather than the identity of this paper.

3 A Service Agreement with Continuation Participation

3.1 Primitives and accepted continuations

At date $t = 0, \dots, T - 1$, a provider observes a public regime z_t and an inherited tier q_t . It chooses stock S_t and a tier $x_t \in [0, 1]$ before demand is realized. Demand D_z is exogenous, nonnegative, and has finite mean. Regime transitions do not depend on the policy. The next inherited tier equals the chosen tier. Premium $r_z x$, indemnity $\kappa x(D_z - S)^+$, and physical cost parameters are fixed in the agreement. There is no private information. With $f_z(S) = \mathbb{E} \min(S, D_z)$ and $L_z(S) = \mathbb{E}(D_z - S)^+$, the customer's expected stage utility is

$$u(z, S, x) = \nu f_z(S) - a(z, S)x, \quad a(z, S) = r_z - \kappa L_z(S), \quad \nu > 0. \quad (1)$$

Maintenance and switching costs are resources expended by the provider, not transfers received by the customer.

The customer may leave *before implementation at every review*. Remaining contractual utility must exceed the publicly specified outside continuation $\underline{U}(h_t)$ at every positive-probability history h_t . The provider commits to its announced contingent protocol; the exit right does not let it retrospectively change prices or tiers. We consider contracts intended to retain the customer at all modeled reviews. A model in which termination is itself an optimized action would need an additional termination decision.

Stock can be contractually specified. It can also be identified from service and cost commitments: if s_z uniquely minimizes $C_z(S) - \xi f_z(S)$, $\xi > 0$, then $F^\pi \geq F^s$ and $C^\pi \leq C^s$ force $S_t = s_{z_t}$ at every reachable review. The nonnegative expected sum of the minimization gaps must be zero. EC Section EC.1 gives the boundary, vector, and relaxed-commitment statements. Thus the tier reduction need not assume away a stock decision that the accepted physical commitments already determine. Without such identification, stock must remain in the accepted optimization.

On a finite exogenous tree, let $w_n = \beta^{t(n)} \Pr(n) > 0$, $c_n = w_n a(z_n, s_{z_n})$, and $i \succeq n$ mean that i is in the subtree rooted at n . Multiplying conditional participation by w_n gives

$$\sum_{i \succeq n} c_i x_i \leq b_n, \quad b_n = \sum_{i \succeq n} w_i \nu f_{z_i}(s_{z_i}) - w_n \underline{U}(n). \quad (2)$$

This derivation imposes every continuation constraint directly. Ex ante participation alone is the special case retaining only the root row. It does not imply the nonroot inequalities. A fixed root payment, when stipulated in the agreement, gives $\sum_i c_i x_i = b_0$; it is a separate payment commitment, not an equality inferred from participation. General formulations below also allow a root inequality.

3.2 Optimized restrictions and switching

Write the accepted set, including the tier box, as

$$X_b = \{x \in C : Ax \leq b, Ex = e\}, \quad C = [\ell, u], \quad W_\lambda(x) = p^\top x - \frac{1}{2}x^\top Qx - \lambda T(x), \quad (3)$$

where $Q \succ 0$, $\lambda \geq 0$, and

$$T(x) = \sum_{a=1}^m \kappa_a |(Dx - h)_a|, \quad \kappa_a > 0.$$

The linear reward p includes net receipts and the linear part of convex operating costs, for example a quadratic cost around a service-specific efficient tier. It need not be proportional to the payment vector c . The general quadratic model and its synthetic examples are not asserted to be estimates of the stock institution above.

Rows of A include (2) and any other retained capacity constraints. E contains fixed payment or physical equalities. D can include temporal and cross-service differences; h incorporates an installed initial tier. The vector-service case uses the same stacked notation, and signed payment coefficients are allowed in (3). Only the specialized transfer lemma requires positive scalar coefficients. All accepted sets used below are nonempty and compact. Strong concavity makes each optimum unique.

A contractual restriction is a predeclared equality $Mx = m_0$, and

$$Y_b = X_b \cap \{Mx = m_0\}, \quad K_\lambda(b) = \max_{x \in Y_b} W_\lambda(x), \quad J_\lambda(b) = \max_{x \in X_b} W_\lambda(x), \quad \Delta_\lambda(b) = J_\lambda(b) - K_\lambda(b). \quad (4)$$

The matrices and commitments are the same on both sides. Pooling decisions at histories with the same permitted information gives one important M . The experiments also pool *amendments* $x - z^{\text{legacy}}$; then $m_0 = Mz^{\text{legacy}}$, which is generally different from pooling actual tiers. We keep that distinction explicit.

3.3 Continuation prices and relative transfers

For a comparator $z \in X_b$, put $g = p - Qz$. Convex optimality gives the following retained foundation.

Proposition 3.1 (Accepted balance). *The comparator z is optimal over X_b if and only if there exist $\mu \geq 0$, unrestricted η , $v \in N_C(z)$, and switching tensions s such that*

$$g = A^\top \mu + E^\top \eta + v + D^\top s, \quad \mu_i (A_i z - b_i) = 0, \quad (5)$$

with $s_a = \lambda \kappa_a \text{sign}(Dz - h)_a$ on nonzero differences and $|s_a| \leq \lambda \kappa_a$ on zero differences. If it is not optimal, a feasible sufficiently short direction has strictly positive directional gain. For optimality over Y_b , add $M^\top \xi$ to (5); ξ prices the restriction rather than a customer participation constraint.

Proof. The objective loss is the sum of a differentiable convex quadratic and a finite convex absolute-value function. Its subgradient plus the polyhedral normal cone characterizes a minimum. Polyhedral normals give the displayed multipliers, without a strict-feasibility assumption. The alternative follows from the directional derivative test for a convex function on a polyhedron. $\square$

In the scalar tree specialization, assume $c_n > 0$ and a fixed root payment. For any feasible z , define its continuation slack $\sigma_n = b_n - \sum_{i \geq n} c_i z_i$ at nonroot nodes. Let B have column n equal to -1 at row n and $c_n/c_{\text{pa}(n)}$ at its parent, with zeros elsewhere.

Lemma 3.2 (Benchmark-relative transfer coordinates). *The correspondence*

$$x = z + By, \quad y_n = -c_n^{-1} \sum_{i \geq n} c_i (x_i - z_i) \quad (6)$$

is a bijection between X_b and the relative transfers satisfying $y_n \geq -\sigma_n/c_n$, the original tier box, and every additional inequality or equality after substitution. The restriction $Mx = Mz$ becomes $MBy = 0$. Both directions of the map take linear work in the number of tree nodes.

Subtree telescoping proves the lemma; EC Section EC.2 gives the inverse explicitly. Negative transfers consume pre-existing continuation slack and cannot be discarded. The lemma is a representation, not a theorem about how much an optimized restriction costs. We next use the restriction prices and the slack in a value theorem.

4 The Value Frontier of Releasing a Restriction

Fix b, λ and the optimized comparator $z = \arg \max_{Y_b} W_\lambda$. Let $\omega \in \mathbb{R}_{++}^k$ set the units of permitted deviations from the restriction. Define

$$V(t) = \max\{W_\lambda(x) : x \in X_b, |Mx - m_0| \leq t\omega\}, \quad t \geq 0. \quad (7)$$

Thus $V(0) = K_\lambda(b)$. The release does not relax participation: it permits previously pooled decisions to differ while preserving every accepted constraint. The full value is reached for finite t , because X_b is compact. Scaling a row of M requires the same scaling of its ω ; otherwise it changes the economic release unit.

4.1 An exact release segment and its first bottleneck

To state the explicit segment, fix a regular active cell at z . Collect permanent equalities, active accepted inequalities and tier bounds, and zero switching differences into $Rx = r$. Retain the signs of all nonzero switching differences and absorb their linear cost into $\bar{p}$. Let R and $[R^\top, M^\top]^\top$ have full row rank. Suppose the active accepted-inequality multipliers are strictly positive, zero-edge tensions are strictly within their capacities, inactive constraints and nonzero switching signs have positive clearance, and the restriction multiplier ξ has no zero coordinate. These are local regularity conditions, not assumptions of the global frontier in the theorem. Rows with zero switching coefficient need not be included as kink constraints.

Choose any matrix Z whose columns form a basis for $\ker R$, and set

$$H = Z^\top QZ, \quad S = [MZH^{-1}Z^\top M^\top]^{-1}, \quad P = ZH^{-1}Z^\top M^\top S, \quad d = Pv, \quad v_j = \omega_j \text{sign } \xi_j. \quad (8)$$

These objects are independent of the choice of basis. S is the marginal curvature of releasing the restriction after the currently binding accepted constraints have been enforced.

Theorem 4.1 (Restriction-release frontier). *For (3)–(7), V is continuous, nondecreasing, concave, and piecewise quadratic. Its unique tier policy is continuous and piecewise affine. At almost every $t > 0$,*

$$V'(t) = \omega^\top (\xi^+(t) + \xi^-(t)), \quad V(t) - V(0) = \int_0^t \omega^\top (\xi^+(s) + \xi^-(s)) ds, \quad (9)$$

where $\xi^\pm$ are optimal multipliers on the two release bounds. At differentiability points the displayed directional price is independent of the choice of optimal multipliers.

Under the regular-cell conditions, let t_1 be the first nonnegative time at which an inactive accepted constraint or tier bound binds, a nonzero switching difference reaches zero, an old active-inequality multiplier reaches zero, a zero-edge tension reaches a capacity, or a component of $\xi - tSv$ reaches zero in its original signed direction. Then $t_1 > 0$ and, for $0 \leq t \leq t_1$,

$$x(t) = z + td, \quad V(t) - K_\lambda(b) = t \sum_j \omega_j |\xi_j| - \frac{1}{2} t^2 v^\top S v. \quad (10)$$

Each event time is an explicit linear ratio obtained from the primal and dual paths. In particular, a currently slack continuation row can be the first bottleneck only at

$$t_n = \frac{\sigma_n}{\sum_{i \geq n} c_i d_i} \quad \text{when } \sum_{i \geq n} c_i d_i > 0; \quad (11)$$

otherwise that row consumes no slack along this segment. In transfer coordinates this is exactly the time at which its lower bound $y_n \geq -\sigma_n/c_n$ is reached.

Proof. Mixing policies feasible at two release levels proves concavity; nesting proves monotonicity. Introduce epigraphs for the absolute values. For fixed active inequalities and switching signs the KKT equations are linear in the parameter, so the unique tier solution is affine and its value is quadratic on that cell. There are finitely many active patterns. Dependent equations may be removed and a finite refinement made; no nondegeneracy assumption is needed for this global piecewise description. Compactness and uniqueness give continuity. Optimal multipliers support the concave value and give (9); integration is valid for this continuous piecewise-quadratic function.

For the explicit segment, a prescribed restriction displacement δ has the cell optimum $x = z + P\delta$. Solving the equality-constrained quadratic and using $RP = 0$, $MP = I$, and $P^\top QP = S$ gives the gain $\xi^\top \delta - \delta^\top S\delta/2$. Its maximum over $|\delta| \leq t\omega$ is at $\delta = tv$ while $\xi - tSv$ retains its signed direction. The old active multipliers and kink tensions also evolve affinely. Until one of the listed events, their signs and capacities, the remaining primal inequalities, and the nonzero switching signs all remain valid. They therefore supply the *full* KKT certificate, not just a solution restricted to the cell. Continuity includes the first event. Substitution into a subtree inequality gives (11). The detailed dual path and degeneracy qualifications are in EC Section EC.3. $\square$

Write $a = \sum_j \omega_j |\xi_j|$ and $q = v^\top S v > 0$. For a desired gain $g \in [0, V(t_1) - K]$, the smallest release on this initial segment is

$$t(g) = \frac{a - \sqrt{a^2 - 2qg}}{q}. \quad (12)$$

No accepted improvement can attain this gain with a smaller release, because (10) is the optimized value, not the gain of a selected direction. At later events one changes the active cell and applies the same equality-constrained calculation. The number of cells need not be polynomial; the theorem is an exact structural path, not a universal complexity claim. If a regularity condition fails, (9) and the global frontier remain valid, but the signed-corner formula is replaced by the appropriate small release quadratic program or a degenerate active-set refinement.

4.2 A complete three-node frontier

Consider a root and two children, $c_i = 1$, tiers in $[0, 1]$, root payment $x_0 + x_1 + x_2 = 3/2$, and child caps $x_1, x_2 \leq 3/4$. Let

$$W(x) = x_1 - \frac{1}{2}(x_0^2 + x_1^2 + x_2^2), \quad Y = \{x \in X : x_1 = x_2\}.$$

The optimized restriction is $z = (1/6, 2/3, 2/3)$ with value $5/24$. Releasing $|x_1 - x_2| \leq t$ gives

$$V(t) - K = \begin{cases} t/2 - t^2/4, & 0 \leq t \leq 1/6, \\ -1/48 + 3t/4 - t^2, & 1/6 \leq t \leq 3/8, \\ 23/192, & t \geq 3/8. \end{cases} \quad (13)$$

The first policy segment is $(1/6, 2/3 + t/2, 2/3 - t/2)$. Child 1 has slack $1/12$, consumes it at rate $1/2$, and therefore binds at $t = 1/6$. The second segment is $(t, 3/4, 3/4 - t)$. At $t = 3/8$, the release price becomes zero and the full optimum $(3/8, 3/4, 3/8)$ is reached. Its relative transfer is $(-1/12, 7/24)$; the negative coordinate uses exactly the initially available child-1 slack. Thus the continuation slack determines an actual change in the optimal policy path, rather than only the domain of a coordinate map. EC Section EC.4 supplies exact primal and dual witnesses for both segments.

5 Capacity, Friction, and Optimized Comparator Rents

5.1 Relaxing a continuation can reduce the adaptation premium

Let $b(t) = b^0 + td$, with $d \geq 0$, and suppose $Y_{b(t)}$ is nonempty on the interval considered. Both optima in (4) are reoptimized for each t .

Theorem 5.1 (Comparator-adjusted continuation rents). *The full and restricted values are continuous, nondecreasing, concave, and piecewise quadratic along the budget path. If $\mu_X(t)$ and $\mu_Y(t)$ are optimal accepted-inequality multipliers, then, at almost every t ,*

$$\frac{d}{dt} \Delta_\lambda(b(t)) = d^\top (\mu_X(t) - \mu_Y(t)), \quad \Delta_\lambda(b(v)) - \Delta_\lambda(b(u)) = \int_u^v d^\top (\mu_X - \mu_Y) dt. \quad (14)$$

Consequently, the adaptation premium is nondecreasing along the path if and only if the rent difference is nonnegative almost everywhere. At a degenerate point, the right directional value derivative along d is the minimum of $d^\top \mu$ over that problem's optimal multiplier set; one must compute the two directional derivatives separately, rather than subtract arbitrary selected multipliers.

Proof. Mixing and nesting give the first two value properties. Fixed-matrix parametric quadratic programming gives the piecewise description as in Theorem 4.1. Weak duality makes each accepted multiplier a supporting price for the budget. At differentiability points all such directional prices coincide with the derivative. Subtract and integrate the two value derivatives. A continuous piecewise-quadratic function is nondecreasing precisely when its derivative is nonnegative almost everywhere. The one-sided rule follows by taking the directional derivative of the infimum of affine-in-budget dual objectives; EC Section EC.5 supplies the polyhedral qualification. $\square$

An exact example makes the distinction material. Keep the preceding three-node objective and root payment, but set $x_1 \leq b$, $x_2 \leq 3/4$, with $b \in [1/4, 2/3]$. The time-only restriction remains $x_1 = x_2$. The full and restricted optima are

$$x^X(b) = ((3/2 - b)/2, b, (3/2 - b)/2), \quad x^Y(b) = (3/2 - 2b, b, b),$$

with cap prices $\mu_X = 7/4 - 3b/2$ and $\mu_Y = 4 - 6b$. Both values increase, but

$$\Delta(b) = \frac{9}{16}(2b - 1)^2, \quad \Delta'(b) = \frac{9}{4}(2b - 1). \quad (15)$$

The premium falls to zero as b rises to $1/2$, then rises again. A claim that more participation slack necessarily makes adaptation more valuable would get the sign wrong on half this interval. The relevant scarcity measure is the difference of the optimized contractual rents.

5.2 Friction thresholds when the restriction is reoptimized

For fixed X, Y , let x_λ and z_λ be their optima. A fixed-comparator balance test is not enough when z_λ changes with friction.

Proposition 5.2 (Optimized-comparator friction path). *The adaptation premium is continuously differentiable for $\lambda > 0$, with a right derivative at zero, and*

$$\Delta'(\lambda) = T(z_\lambda) - T(x_\lambda). \quad (16)$$

On every bounded friction interval its zero set is a finite union of closed intervals and isolated points. It can be computed from the restricted parametric policy path without solving a full-class optimization at every friction: refine that path at changes of its accepted active rows and switching signs, and on each resulting cell impose (5) at the affine z_λ . This is a linear feasibility problem jointly in λ and the full-class prices; its projection on λ is an interval, possibly empty or a point. The union of these projections is exactly the zero set. There is no general upward-ray conclusion for arbitrary restrictions.

Proof. Uniqueness and compactness imply continuous optimizers. The envelope argument gives $J'(\lambda) = -T(x_\lambda)$ and $K'(\lambda) = -T(z_\lambda)$, including active-set changes. Both paths are piecewise affine. On a refined cell, every sign in (5) is fixed or zero, z_λ is affine, and tension bounds and stationarity are linear in the displayed variables. The full balance test is necessary and sufficient. A projection of a polyhedron onto a line is an interval. Finitely many cells establish the assertion; the method does not bound their number polynomially. $\square$

For example, use the root payment $3/2$, the box and child caps above, $p = (1/2, 3/4, 1/4)$, $Q = I$, and $T(x) = |x_1 - x_0| + |x_2 - x_0|$. A restriction $x_1 - x_2 = 1/2$ has optimized policy $z = p$ for every λ : its switching cost is already the smallest possible under that restriction. The full optimum is $(1/2, 3/4 - \lambda, 1/4 + \lambda)$ until $\lambda = 1/4$, and is constant thereafter. Therefore

$$\Delta(\lambda) = \begin{cases} \lambda^2, & 0 \leq \lambda \leq 1/4, \\ \lambda/2 - 1/16, & \lambda \geq 1/4. \end{cases} \quad (17)$$

More friction increases the value of removing a restriction that forces switching. This does not contradict constant-comparator critical-friction results: their restriction and zero-switching benchmark are different.

5.3 Which information expansion creates the value?

Let $Y_0 \subseteq \dots \subseteq Y_k = X$ be nonempty predeclared classes and $K_j = \max_{Y_j} W$. Then

$$J - K_0 = \sum_{j=1}^k (K_j - K_{j-1}), \quad K_j - K_{j-1} \geq 0. \quad (18)$$

The identity is elementary; its economic use requires each K_j to be optimized under the *same* accepted constraints. A release path between adjacent equality-defined classes prices that particular increment through Theorem 4.1. The ordering matters when information features interact; the increments are not automatically Shapley values or unique causal attributions.

The fixed class describes an unchanging amendment or tier. The time-only class describes a preannounced schedule. The current-regime class requires public regime measurement and a date–regime decision table. A two-period-memory class additionally requires retaining the preceding regime. The full-history class has no such information pooling. These are operational restrictions, not an assertion that time-only is the universal status quo. Current regime alone can be insufficient even when exogenous primitives are Markov, because the inherited tier and the outstanding promise are endogenous state variables. The next section establishes the corresponding sufficient-state formulation.

6 Promised-Payment Recursion and Horizon Complexity

Suppose the regime has a finite public Markov transition matrix P_t , the one-period net payment is $a_t(z)x$, and the reward is $g_t(z, q, x)$. Suppose further that each conditional payment cap $\bar{b}_t(z)$ and the feasible

actions depend only on the displayed state and date. A general history-dependent outside option does not satisfy this assumption merely because demand is Markov.

Let $\mathcal{V}_t(z, q, b)$ denote the maximum conditional reward from date t with an *exact* remaining discounted payment promise b . Put $\mathcal{V}_T(z, q, 0) = 0$ and $\mathcal{V}_T(z, q, b) = -\infty$ for $b \neq 0$. For $b \leq \bar{b}_t(z)$, define

$$\begin{aligned} \mathcal{V}_t(z, q, b) &= \max_{x, \{b_{z'}\}} g_t(z, q, x) + \beta \sum_{z'} P_t(z, z') \mathcal{V}_{t+1}(z', x, b_{z'}) \\ \text{subject to} \quad &a_t(z)x + \beta \sum_{z'} P_t(z, z') b_{z'} = b, \\ &b_{z'} \leq \bar{b}_{t+1}(z') \quad \text{for positive-probability children.} \end{aligned} \tag{19}$$

Outside this cap, set the value to $-\infty$. A zero-probability successor imposes no obligation. For a root upper budget rather than an equality, maximize the root value also over permitted b .

Theorem 6.1 (Exact sufficient-state implementation). *Under the stated Markov sufficiency and compact-attainment conditions, (19) has the same value as the full accepted history-tree problem with its specified root promise. There exists an optimal deterministic contract implemented using only (t, z, q, b) . Conversely, a feasible recursion policy induces all tree continuation constraints. With several additive commitments, the result holds with a vector of remaining commitments. A history-dependent cap, nonadditive global resource, or omitted physical state must be included in the state or treated on the full tree; otherwise the claimed equivalence need not hold.*

Proof. For any history policy, take its conditional remaining payments in each child subtree as the child promises. They satisfy the equality and all child caps, and induction bounds its conditional reward by the right side. Conversely, combine an attained current maximizer with child maximizers at the allocated promises. Induction gives every child obligation, while the displayed equality gives the parent's exact promise. This proves equality and constructs a state-based policy. The vector case applies the same induction to each additive commitment. $\square$

Conditional normalization is essential. Multiplying the conditional budget at node n by w_n recovers (2). Within (19), the child weights are $\beta P_t(z, z')$, not unconditional probabilities multiplied a second time. Promises are decision variables allocated across next-period regimes; they are not forecasts fixed independently of the current decision.

A branching factor B and T decision dates can create $N = (B^T - 1)/(B - 1)$ history nodes. The linear transfer map is linear in N , not in T . The recursion avoids materializing those histories when the displayed state is sufficient, but its promise state is continuous in the continuous-tier model. A grid with G promise points, Q_0 inherited-tier points, S_0 regimes, and k commitment coordinates has order $TS_0Q_0G^k$ value entries; choosing child allocations is an additional cost and can grow exponentially in the number of successors. Such a grid is an approximation unless the contract institution itself has the stated finite actions and attainable promises. Our exact recursion checks use the latter finite-action interpretation, and exhaustively compare every attainable root promise with the full history enumeration. No interpolation theorem or unrestricted polynomial-horizon guarantee is inferred from them.

7 Implemented Gains and Reusable Comparator Certificates

A feasible implemented $\hat{x}$ and a valid upper bound $U_Y \geq K$ give the economic certificate

$$W(\hat{x}) - K \geq W(\hat{x}) - U_Y. \quad (20)$$

An upper bound $U_X \geq J$ separately gives $J - W(\hat{x}) \leq U_X - W(\hat{x})$. The first establishes an improvement over an optimized restriction; the second establishes implementation accuracy. Neither follows from predictor error or an optimizer's termination label. EC Section EC.6 states the box-conjugate bound used by the independent verifier, retaining every participation row, equality, and switching term.

7.1 A comparator envelope requiring no new query solve

Consider a fixed restricted geometry Y and fixed convex cost ϕ with strong-convexity matrix $D_0 \succ 0$:

$$K(h) = \max_{x \in Y} \{(b + U^\top h)^\top x - \phi(x)\}.$$

Let the columns of Z_Y span a linear space containing every difference of two points of Y , and choose

$$L_Y \geq \|UZ_Y(Z_Y^\top D_0 Z_Y)^{-1/2}\|^2.$$

A trace bound on the corresponding positive semidefinite matrix is sufficient and can be evaluated rationally.

Proposition 7.1 (Certified reusable restricted-value envelope). *At an anchor h_i , let $z_i \in Y$ and $U_i \geq K(h_i)$ be certified, and put $\delta_i = U_i - W_{h_i}(z_i)$. Then, for every reward query h ,*

$$K(h) \leq U_i + (U z_i)^\top (h - h_i) + \sqrt{2L_Y \delta_i} \|h - h_i\| + \frac{1}{2} L_Y \|h - h_i\|^2. \quad (21)$$

The minimum of these bounds over a finite cache remains valid. After storing U_i , $U z_i$, δ_i , and h_i , each query uses order r arithmetic operations per anchor, where r is the reward dimension, and no restricted optimization. The bound is valid even when the anchor decision is not exactly optimal.

Proof. Strong convexity gives $\|z_i - x^*(h_i)\|_{D_0}^2 \leq 2\delta_i$. Restricting differences to the displayed linear space bounds the gradient error by $\|U(z_i - x^*)\| \leq \sqrt{2L_Y \delta_i}$. The same strong-monotonicity argument gives an L_Y -Lipschitz gradient $\nabla K(h) = U x^*(h)$. Apply the smooth upper Taylor bound, substitute $U_i \geq K(h_i)$, and bound the anchor-gradient error by Cauchy-Schwarz. EC Section EC.7 gives the full argument, including exact outward square-root rounding. $\square$

This is a nearby-reward reuse mechanism, not a theorem for an arbitrarily changing accepted geometry. Changing continuation matrices, budgets, switching coefficients, or information restrictions requires a new valid cache or a bound that explicitly covers those changes. A loose envelope is still valid but may not certify a positive economic gain. A deployment can then request a new comparator certificate; that solve must be charged. Accepted-policy implementation itself need not perform an economic-comparator batch on every query.

7.2 Why the robust outer comparator is locally continuous

Write a designed family as

$$Y_\theta(\rho) = \{x \in C_0 : (A^0 + \rho A_\theta)x \leq b^0 + \rho b_\theta\},$$

where C_0 is the common compact box and equality-restricted class, and θ belongs to a compact uncertainty set. Let η_i bound $\sup_{\theta, x \in C_0} |(A_\theta)_i x - (b_\theta)_i|$ and put $O(\rho) = \{x \in C_0 : A_i^0 x \leq b_i^0 + \rho \eta_i\}$. This radius-dependent outer class contains every $Y_\theta(\rho)$ and equals the nominal class at zero. It is not a fixed large union bound switched on at every positive radius.

Proposition 7.2 (Local outer-comparator loss). *Suppose an anchor in C_0 has slack at least $\gamma_i > 0$ in every perturbed row for $0 \leq \rho \leq \rho_0$. Suppose the objective W_θ is uniformly L_W -Lipschitz on C_0 , whose diameter is D_C . Then*

$$0 \leq \max_{O(\rho)} W_\theta - \max_{Y_\theta(\rho)} W_\theta \leq L_W D_C \max_i \frac{2\rho\eta_i}{\gamma_i + 2\rho\eta_i}. \quad (22)$$

Thus the outer-set component of certificate slack vanishes at least linearly as the radius tends to zero under these assumptions. The constant can be large when an important accepted row has little strict slack.

Proof. An outer-feasible decision violates any true row by at most $2\rho\eta_i$. Mix it with the anchor by the largest fraction in (22). Every resulting true row is feasible, and the common equalities and box remain feasible by convexity. The movement is at most that fraction times D_C . Apply the Lipschitz bound to an outer maximizer. $\square$

The row envelope can be tightened by taking its support over the equality-restricted box rather than an unrestricted box. EC Section EC.8 also gives a valid successive-outer-set refinement. Neither a small uncertainty radius nor set containment alone establishes field robustness: the uncertainty set remains a designed, uncalibrated family. Without a uniform anchor, the linear rate above is not asserted. The finer-radius experiment distinguishes high local sensitivity from an actual positive-radius discontinuity.

8 Computational Evidence

8.1 Protocol and common-instance comparator hierarchy

We use two explicitly different families. The first takes the first 24 nominal contexts of the frozen R24 design: 63 history nodes, two service coordinates, six resource-reward coordinates, all continuation rows, and four shared capacities. A pooled action is an equal *amendment* to the existing heterogeneous tier. The fixed-amendment class is the singleton zero amendment because the exact root payment prevents a nonzero uniform change. Calling this an optimized constant actual tier would be incorrect.

The second family has 63 nodes, one tier, a common baseline of $1/2$, an equiprobable binary public regime, and Markov stage rewards, curvature, and conditional continuation caps. All five classes are feasible and pool actual tiers. The exact root commitment fixes the optimized constant tier at $1/2$ but leaves time-only schedules nontrivial. Every adjacent comparison uses the same primitive instance. We vary reward and

switching parameters across 24 seeded contexts; EC Section EC.9 gives the integer and rational design. Integer probability weights multiply all conditional weights by 32, so its raw value unit differs from the legacy family. We report shares within each family, not a cardinal comparison between the two families.

Table 1 reports the optimized increments. Full-history amendments retain substantial value beyond two-period memory on the legacy family. In the public-Markov family most of the gain comes from observing the current regime; the additional memory and full-history gains are smaller and can vanish within the numerical bracket on particular instances. This distinction is informative: the total legacy gain is not presented as a uniquely contractual effect or attributed entirely to learning. Regime-only controls omit inherited tiers and promises even in the Markov family. The exact augmented-state result remains Theorem 6.1.

Each optimized class has a rational feasible policy and a separate rational dual upper bound. If these are $[L_j, U_j]$, the certified interval for an increment is $[\max\{0, L_j - U_{j-1}\}, U_j - L_{j-1}]$. We do not interpret a tiny difference between two approximate lower endpoints as a negative true increment. Both the lower-endpoint summaries and these enclosing intervals are retained in the raw record. Exact symbolic tests independently verify the three-node release path, the capacity rent example, and the optimized friction example. Additional rational matrix tests verify the regular-cell release formula. Finite-action promise recursions match exhaustive history enumeration at every attainable root promise; these finite checks supplement the induction proof rather than replace it.

8.2 Radius refinement and comparator reuse

Table 2 recomputes the true and common-outer time-only comparators on eight fixed legacy contexts and ten radii, including much smaller positive radii than the earlier study. Each hidden coefficient vector is held fixed while its radius varies. The outer-set slack tends toward zero and rises steeply before the earlier smallest positive radius. Thus the coarse positive-radius plateau did not establish a discontinuity. The construction already depended continuously on radius; Proposition 7.2 explains its local behavior. This experiment isolates the outer-set contribution and does not conflate it with objective interpolation over uncertainty vertices.

For comparator reuse we fix the legacy time-only geometry and switching coefficient, build four certified anchors, and validate 32 nearby reward queries. The rational trace curvature bound is approximately 0.45830. The largest envelope-minus-certified-lower-value gap is reported in Table 3. The deployed envelope makes no query optimization calls; the 32 validation solves are offline work used to measure tightness. The four initial solves and storage are not free. This is a test of a reusable upper bound and its quality, not a measured end-to-end speedup over the historical robust pipeline.

8.3 Historical learning and timing evidence

The R24 matched study charged optimizer labels, fitting, prediction, auditing, polishing, and fallback against lifted continuation. At the strict target both learned routes fell back in every measured case at 63, 127, and 255 nodes. Their mean total times exceeded the lifted comparator in every configuration, with unfavorable paired intervals. The coarse target also produced no observed finite amortization case. Table 4 retains

the strict-target comparisons; the complete phase-resolved record and all less favorable and more favorable individual cases remain in the code/data archive. These are implementation diagnostics, not a second positive contribution about learned optimization.

Let $C_{\text{off}} \geq 0$ be a candidate’s incremental offline cost, and let t_C, t_L be its and the lifted method’s per-query times. The candidate-minus-lifted total cost after Q_0 queries is

$$C_{\text{off}} + Q_0(t_C - t_L) = C_{\text{off}} - Q_0\delta, \quad \delta = t_L - t_C. \quad (23)$$

For positive offline cost a finite positive crossing exists only when $t_C < t_L$, at $Q_0 = C_{\text{off}}/(t_L - t_C)$. Strict savings require a query count above this crossing. The analysis code uses this convention, and the displayed equation now matches it. The historical observations have no positive learned crossing under that protocol.

The historical robust study compared different achieved gains. Its roughly 20-millisecond differences between protected learned policies and robust maximin are therefore descriptive cost accounting, not matched-quality dominance. We do not use these three isolated method points as a computational advantage claim. Their complete gain–cost coordinates remain in the archive. The new comparator envelope addresses a different bottleneck under narrower, explicitly fixed-geometry conditions; it is not retroactively substituted into the old timings.

9 Conclusions

An accepted adaptation premium is a difference between optimized contract values. The release frontier identifies the gain from relaxing an information restriction, its curvature, and the first continuation slack that limits the improvement. The comparator-adjusted rent decomposition explains why a capacity relaxation can lower that premium and why greater friction can increase it. These statements depend on the actual restriction, not on a presumption that more capacity or less friction has the same effect on both optimized classes.

The public-state institution also separates contractual memory from the exogenous regime. A remaining-payment promise and inherited tier can make a history policy recursively implementable, but the resulting continuous state still carries a computational cost. The numerical hierarchy shows which restrictions matter in each designed family instead of assigning all positive value to a convenient time-only benchmark. Independent policy and comparator certificates then connect these structural values to implemented decisions. Learning remains one possible implementation route; the tested protocol provides no evidence that it is the cheapest one.

Data and Code Accessibility

The separate code/data package contains the R25 source, rational certificate records, exact example checks, finite-action recursion checks, replay and build scripts, a point-by-point response, and a preservation manifest. It also contains the complete predecessor manuscript, formal companion, computational record, historical supplement, and earlier repository derivations. None of the predecessor mathematical statements or empirical

tables is silently discarded. The current manuscript and its formal electronic companion form the journal-facing argument; the predecessor and computational documents are an indexed research archive rather than additional required theorem layers. No field data are used. Repository delivery is for further author/referee review and is not a submission to the journal.

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Table 1: Value of progressively richer optimized contracts

Expansion	Legacy amendments		Markov service tiers	
	Mean gain	Share (%)	Mean gain	Share (%)
Fixed to time-only	0.116155	7.95	0.127557	0.49
Time-only to regime	0.136312	9.33	26.066328	99.34
Regime to two-period memory	0.224350	15.36	0.028788	0.11
Two-period memory to full history	0.983916	67.36	0.016938	0.06
Total	1.460733	100.00	26.239611	100.00

There are 24 common instances per family and five optimized classes per instance. Means use certified lower endpoints; the complete enclosing increment intervals are in the data. Maximum optimization-bracket widths are $8.13\text{e-}08$ and $2.97\text{e-}07$, respectively. Shares divide each mean increment by the within-family total. Raw value units differ between families. Current regime omits inherited tier and remaining promise.

Table 2: The common outer comparator becomes locally tight

Uncertainty radius	Mean upper bound on outer-set slack
0	0.000000003
1/65536	0.000047365
1/16384	0.000189415
1/4096	0.000757072
1/1024	0.003019002
1/256	0.011836705
1/64	0.045064142
1/16	0.073768497
1/8	0.074890218
1/4	0.078860565

Eight fixed contexts and ten radii, with a directly optimized true and outer comparator at each pair (160 certificates). Entries average the outer certified upper value minus the true certified lower value. At zero the true outer-set slack is exactly zero; the displayed residual is the optimization bracket. Maximum individual bracket width is $3.06\text{e-}08$. This isolates outer-set loss and is not a calibrated uncertainty experiment.

Table 3: Restricted-comparator reuse with fixed geometry

Quantity	Value
Certified anchors	4
Nearby reward queries	32
Restricted optimizations in deployed query evaluation	0
Query optimizations used only for offline validation	32
Curvature trace bound	0.458298401
Mean envelope slack upper bound	0.000129719
Maximum envelope slack upper bound	0.000248488

A common time-only feasible set and switching coefficient are held fixed. Each query evaluates all four analytical anchor envelopes and takes their minimum. The slack is cached upper value minus independently certified query lower value; it therefore encloses the actual upper-bound loss. Anchor construction, policy acceptance checks, storage, and offline validation are separate costs. No end-to-end timing advantage is inferred.

Table 4: Preserved strict-target end-to-end timing comparisons

Nodes	Lifted (ms)	Tanh (ms)	RBF (ms)
63	10.34	18.84	18.91
127	96.24	112.53	113.62
255	239.93	277.86	274.96

Historical R24 measurements, unchanged. Target 10^{-5} ; 32 held-out contexts and three repetitions per method and size. Both learned routes have 100% fallback at each size. Their paired saving intervals versus lifted continuation are entirely negative. Offline label/fitting costs are retained separately and cannot turn negative online savings into amortization. These are not newly timed R25 measurements.